



Comparative analysis of primers and alternative polypropylene pre-treatment techniques

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ABSTRACT

The test polypropylene substrates obtained from the original plastic car parts for Škoda Superb were subjected to 10 pre-treatment techniques: solvent cleaning (isopropanol, xylene), mechanical abrasion, immersion in a chromic acid mixture, flame activation, atmospheric and low-pressure plasma treatment, corona discharges and the application of two different primers (also called adhesion promoters). One of the primers, based on chlorinated polypropylene, was commercially available, while the other was prepared in our laboratory using polypropylene grafted with maleic anhydride. The test substrates were characterized using contact angle measurement for designation of surface free energy, Atomic Force Microscope (AFM) for roughness analysis, Scanning Electron Microscopy (SEM) for surface topography, and Energy Dispersive X-ray Spectrometry (EDX) for surface oxygen and contamination analysis. The suitability of the surface preparation methods in the industry was evaluated through peel strength tests (adhesive bonding process) and cross-cut tests (painting process). Not all tested methods were effective in increasing adhesion, especially in the adhesive bonding process, and even fewer in the painting process. However, some of them were suitable for both applications. The use of primers was found to be crucial.

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1. Introduction

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Surface activation is still an important topic in various industries.^[1–4] It mainly concerns polyolefins, polymers with low surface free energy built only with carbon and hydrogen atoms. This group includes polypropylene (PP), which combines the best features of various materials: low density, ease of processing, recyclability, high mechanical endurance, and cost-effectiveness.^[5,6] Due to the hydrophobic nature of this polymer and the lack of functional groups, it exhibits high chemical resistance. However, this characteristic poses difficulties in surface treatment such as printing, painting, gluing, or electroplating.^[7,8]

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Several different surface activation methods are employed in the pre-treatment processes.^[9–11] Among them, the most popular are automatic systems: flame activation, corona discharge, and atmospheric or low-pressure plasma.^[12–16] These methods significantly enhance the surface activity of polyolefins, offer process repeatability, and require minimal processing time. However, the acquisition and operation of these devices involve substantial capital expenditure. Additionally, setting and adjusting the initial parameters during the process can be complicated and time-consuming.

Moreover, the lifespan of the machines is limited, and they are dedicated only to specific projects. Any interference in the process necessitates costly modifications to the entire apparatus. Furthermore, changes in environmental conditions (such as increased air humidity or temperature) or installation parameters (such as voltage surge) may damage modified surface or alter its wettability. As a result of applying these methods, a layer of low molecular weight oxidized materials (LMWOM) soluble in water appears on the substrate. This layer can negatively affect bonding and paint adhesion and lead to the hydrophobic recovery effect.^[17]

On the other hand, other methods are also popular in the industry, such as solvent cleaning, mechanical abrasion, applying primers, and wiping in a chromic acid mixture.^[18,19] These techniques are primarily utilized by small and medium-sized companies engaged in low-scale production. With limited budgets, these methods allow for achieving a similar surface activation effect as automatic methods. Moreover, they can be employed manually or in combination with other techniques.^[20,21]

Solvent cleaning, apart from removing surface contaminants, can induce partial swelling and alter the crystallinity of polymer surfaces, thereby influencing mechanical and diffusion adhesion.^[22,23] Mechanical abrasion increases contact surface area by creating pores or leveling the material. Encinas et al.^[7] indicated that mechanical abrasion led to a slight increase (approximately 2%) in the polar component of surface free energy and static friction coefficient of low-density polyethylene (LDPE).

Immersion in a chromium mixture was a popular technique due to the quick and effective enhancement of adhesive properties of polyolefins. However, it poses hazards to workers, results in significant environmental burdens, and alters the appearance of activated objects after application.

Of particular interest is the primers technique, which has gained popularity in the automotive industry.^[24,25] This method involves applying a small amount of polymer solution, forming an adhesive layer between the substrate and the coating after the solvent evaporates. The connection between the polypropylene substrate and the primer is established through the diffusion of polymer chains, which interlock, creating a stable and durable interphase between the materials. On the other hand, the joint between the primer and the applied adhesive/paint coating relies on Van der Waals interactions of

polar functional groups present on both surfaces.^[26,27] Primers not only enhance adhesion properties but also significantly improve the resistance of the adhesive joint to organic solvents (such as gasoline), humidity, and temperature conditions.^[28] 85

The primary constituent of polyolefin primers is chlorinated polyolefins, which readily penetrate the substrate and ensure proper wetting.^[29,30] However, the presence of chlorine molecules in both the production and application stages renders this polymer harmful to humans and the environment.^[31,32] Introducing primers based on polypropylene functionalized with other polar monomers eliminates toxic chlorine while maintaining good adhesion properties. One such monomer, maleic anhydride, effectively addresses the issue of chlorine presence.^[33–35] 90

Several researchers have compared different methods for surface activation of polyolefins.^[36–40] Strobel et al.'s research^[36] demonstrates that flame activation yields inferior results compared to the corona discharge technique, and excessive flame treatment can lead to surface degradation and matting. However, with careful selection of process parameters, flame treatment can effectively compete with chemical processing techniques, including activation in a chromium mixture.^[37] Wingfield^[38] extensively compared multiple polypropylene activation techniques by bonding samples with epoxy adhesive. The strength of the joints decreased in the following order: application of primers > corona discharges > immersion in a chromium mixture > mechanical abrasion > solvent washing. Kostov et al.'s research^[39] indicates that the atmospheric plasma activation technique may be more effective than corona discharges for PP substrates, reducing the contact angle from 93° to 36° and 49°, respectively, for the two techniques. In another study,^[40] low-pressure plasma treatment was compared with activation in a chromium mixture on PE surfaces. Using the chemical method, the samples bonded with epoxy resin exhibited lower peel strength (0.7–1.7 MPa) than those treated with O₂ plasma (1.8–4.4 MPa). Nevertheless, there is still a shortage of literature summarizing the performance of various surface activation methods. 95 100 105

This research aims to compare the effectiveness of primer application against other surface activation treatment techniques. The methods will be assessed in terms of wettability, surface contamination levels, roughness, and two commonly used industry tests for evaluating surface activation effectiveness: the ISO 2409 (cross-cut)^[41] cut test for the painting process and the ASTM D093–98 for peel strength measurement (peel 180).^[42] 110 115 120

Table 1. Material properties of 20% talc filled polypropylene used in studies.

Typical properties	Value	Test Method
Melt Flow Rate (230°C/2.16 kg)	18 g/10 min	ISO 1133-1
Density, (23°C)	1.05 g/cm ³	ISO 1183-1/A
Flexural Modulus, (23°C)	1900 MPa	ISO 178/A1
Tensile Stress at Yield, (23°C)	22 MPa	ISO 527-1, -2
Vicat Softening Temperature	136°C	ISO 306

2. Materials and methods

2.1. Materials

The following materials were used in this study:

The substrates were made of 20% talc filled polypropylene Hostacom TRC 352N E1 C11306 (Basell Orlen Polyolefins, Poland). Material properties obtained from the supplier are listed in Table 1.

Adhesion Promoter 4298UV (3 M, Poland) was used as the primer. Isopropanol 99.7% and xylene (mixture of isomers) 98.5% were used as solvents. Sodium dichromate Na₂Cr₂O₇ · 2 H₂O 99% (Polish Chemical Reagents, Poland) and sulphuric acid 98% (Alchem, Poland) were used in the chromic acid mixture.

Benecke-Kaliko Automobile Interior Color Decklack Titanschwarz (Benecke-Kaliko, Poland) in the form of a spray was used for the cross-cut test according to ISO 2409.^[41]

Double-sided tape DuploCOLL 56,066 (Lohmann Poland), polyethylene foam coated with acrylic glue with a shear strength of 50N/625 mm² and a peel strength of 25N/25 mm at 23°C was used for the peel 180 strength test according to ASTM D903-98.^[42]

All the parts involved in the adhesive joint (substrate, adhesive tape, primer, paint) are commonly used in the automotive industry.

2.2. Sample preparation

Maflow Plastics Poland provided the test substrates from the original car parts for Škoda Superb, specifically from the boot section.

The car parts were manufactured using an injection molding machine, Toshiba IS 3500 DF (Toshiba Machine CO LTD.), using 20% talc-filled polypropylene Hostacom TRC 352N E1 C11306. The injection temperatures for the different zones averaged 229 ± 7°C. The hot channel system in the mold was maintained at a temperature of 260°C. As per VW classification, the resulting parts exhibited a K3N car texture characterized by fine graining.

In the next step, the test substrates were cut into 20 × 203 × 2 mm strips from the molded parts, following the ASTM D903-98.^[42] These strips were then used without pre-cleaning after the injection process. Each cut strip was

Table 2. Description of used surface activation methods.

Designation	Methods	Description
O	Not-treated sample	Without surface treatment.
I	Isopropanol cleaning	Washing for 15 minutes.
K	Xylene cleaning	Washing for 15 minutes.
H	Mechanical abrasion	Emery sponge; grit 500; $t=60$ s.
Cr	Chromic treatment	Mixture $\text{Na}_2\text{Cr}_2\text{O}_7$: H_2SO_4 : H_2O (1 : 17.5 : 1.2 wt.); $t=15$ min; $T=65^\circ\text{C}$.
F	Flame treatment	Automatic line Teca-Print (Switzerland); Fuel – propane 100%; $t=0.2$ s; gas flow 1.2 l/min; air flow 25 l/min.
PA	Cold atmospheric plasma	Atmospheric oxygen plasma device PLS300 (ELTO, Poland). Discharges width 20 mm; $P=1000\text{W}$; conveyor speed 2.8 m/min.
LP	Low pressure plasma discharges	$P=100\text{W}$; $t=90$ s.
CD	Corona discharges	Test stand equipped with quartz electrodes to metals and non-metals activation. Power generator $U_1=230\text{V}$; $P=700\text{W}$; conveyor speed 2.8 m/min.
P	Applying primer layer (commercially available)	Applying one primer layer Adhesion Promoter 4298UV by using a paintbrush.
Pm	Applying primer layer (own formulation)	Applying one primer layer (own formulation based on polypropylene grafted maleic anhydride) by using a paintbrush.

subjected to one of the ten pre-treatment techniques described in [section 2.3](#). The activated surfaces were analyzed using the methods outlined in [section 2.4](#). 155

2.3. Pre-treatment methods

All ten pre-treatment methods described in [Table 2](#) are commonly used in the industry for surface activation.

2.4. Research methods 160

2.4.1. Contact angle measurement

The contact angle (θ) measurements were conducted at room temperature using two fluids: diiodomethane (a non-polar liquid) and water (a polar liquid). The measurements were performed using a DSA G10 goniometer with a drop shape analysis system (Krüss GmbH, Germany). The contact angle values reported were the average of 10 measurements. 165

The Owens-Wendt method was employed to calculate the surface free energy (SFE) and its polar and dispersion components.^[43] According to this method, the surface free energy of the solid surface (γ_s) is the sum of its polar (γ_s^p) and dispersive (γ_s^d) components, as expressed in Equation (1): 170

$$\gamma_s = \gamma_s^p + \gamma_s^d \quad (1)$$

The polar component (γ_s^p) accounts for polar, hydrogen, inductive, acidic, and basic interactions, while the dispersive component (γ_s^d) characterizes London's dispersive forces.

For the measurement, two liquids with known polar (γ_l^p) and dispersive (γ_l^d) component values were required, along with two equations, one for each liquid. The equations are formulated as follows (2): 175

$$0,5\gamma_l(1 + \cos \theta) = (\gamma_s^d + \gamma_l^d)^{0,5} + (\gamma_s^p + \gamma_l^p)^{0,5} \quad (2)$$

By solving the system of equations, the values of γ_s^d and γ_s^p for the solid surface can be determined and used in equation (1).

2.4.2. Energy dispersive X-ray spectrometry (EDX) and Scanning Electron Microscopy (SEM) analysis 180

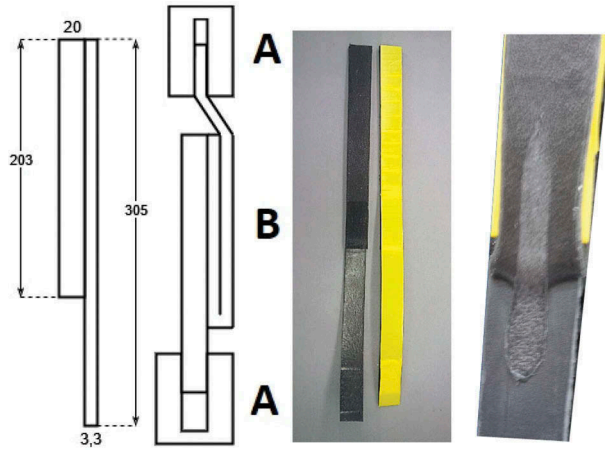
The analyses were conducted on 1 cm² squares obtained from the activated samples. The studies were performed using Scanning Electron Microscopy (SEM) with a secondary electron detector (SE) on a LEO Electron Microscopy Ltd model 1430 VP (LEO Electron Microscopy Ltd, United Kingdom). The SEM analyses were performed at a low vacuum of 50 Pa and using an accelerating voltage of 20kV, which allowed us to avoid sputtering the gold layer. EDX (Energy Dispersive X-ray Spectrometry) studies were managed using an X-ray spectrometer, Quantax 200, equipped with an XFlash 4010 detector (Bruker AXS, Germany). EDX spectrometry facilitated the quantitative analysis of the elemental composition of the surfaces. This technique allowed for assessing the levels of oxygen and contaminants present after each surface pre-treatment method. The results were averaged from two locations on the analyzed surface. 185 190

2.4.3. Atomic force microscopy (AFM) 200

The atomic force microscopy (AFM) studies were carried out using 1 cm² pieces of activated substrates. Images were acquired using a MultiMode NanoScope IIIa (VeecoMetrology Inc., USA) equipped with a silicone scanning tip (Veeco, USA). The AFM operated in tapping mode in ambient conditions at room temperature. In this paper, the term R_{max} refers to the maximum height observed within the scanned area of 10 μm^2 on the surface sample. Values were averaged from at least five locations on the analyzed surface. 195 200

2.4.4. Mechanical properties – peel 180 strength 205

The activated polypropylene strips were bonded with the DuploCOLL 56,066 double-sided adhesive tape using a hydraulic press (Bego Hydrofix, Germany). A pressure of approximately 20 bar was applied during the bonding process, and the pressing time was set to 60 seconds. Mechanical property tests were performed at room temperature after 7 days of adhesive joint conditioning using a TIRA test 27,025 testing machine (TIRA Maschinenbau GmbH, Germany). The measurements were carried out according to ASTM D903–98 standards, with a testing velocity of 152 mm/min (Figure 1). Three repetitions of strength tests were performed for each activation method. 210



Q4 **Figure 1.** Left side - dimensions and setting of test sample (b) in endurance machine grip (a) according to ASTM D903-98.^[42] right side - appearance of sample before and during the test.

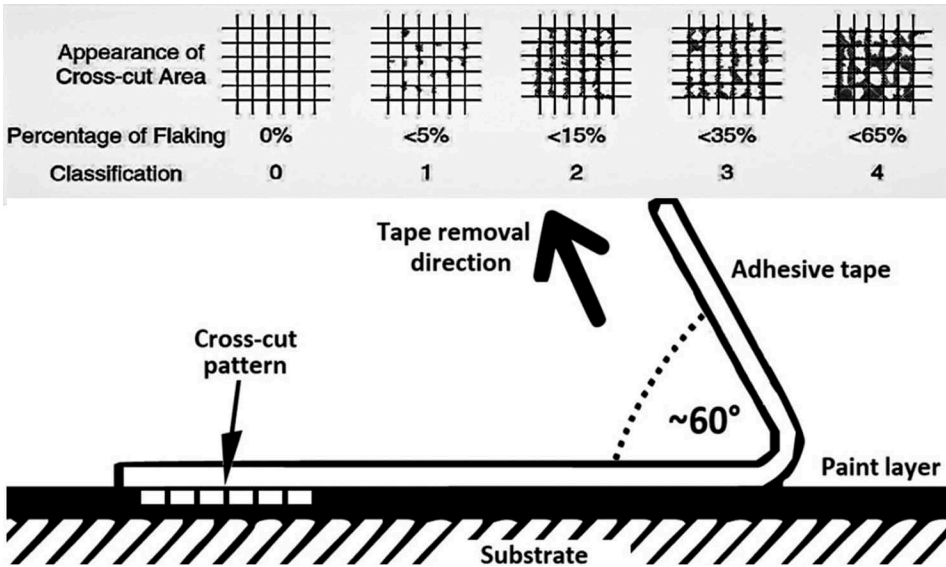


Figure 2. Classification of percentage of flaking acc. To ISO 2409 (above),^[41] removal of adhesive tape direction (below).

2.4.5. Test of adhesion of the paint film – cross-cut test

The activated substrates were coated with a single Benecke-Kaliko Automobile Interior Color paint layer. A cross-cut pattern was performed using a Peters knife, and then a 25 mm wide adhesive tape was applied to the damaged area according to ISO 2409. The adhesive tape was peeled off quickly in the direction shown in Figure 2. After removing the tape, the amount of paint adhered to the tape was examined and evaluated following the mentioned standard. Image Color Summarizer^[44] was employed to determine the amount of color remaining on the tape after peeling off.

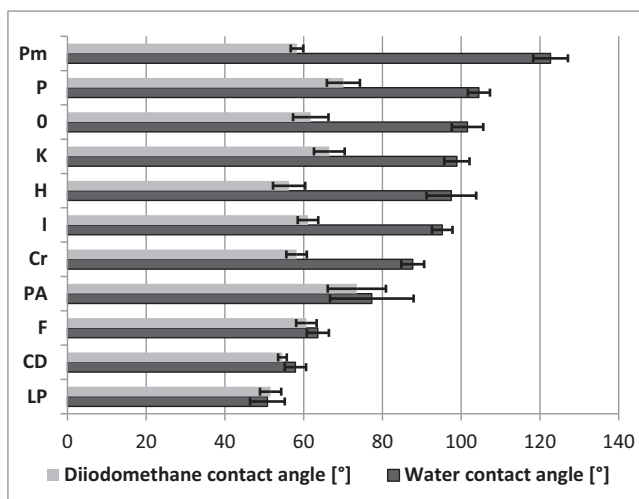
3. Results and discussion

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3.1. Contact angle measurement

After applying each pre-treatment technique, contact angle measurements were conducted using water and diiodomethane. The surface free energy components (γ_s), including the polar (γ_s^p) and dispersion (γ_s^d) components, were calculated by using the Owens-Wendt method. **Figure 3** presents the average values obtained from at least 8 measurements of polar and non-polar liquid contact angles.

Non-activated polypropylene substrates typically exhibit low surface free energy due to the absence of oxygen groups on the surface. These oxygen groups are introduced into the polymer structure through high-energy absorption, which can break the C-C bonds. Techniques involving automatic devices that generate electrical discharges, such as corona discharges (CD), atmospheric plasma (PA), flame treatment (F), and low-pressure plasma (LP), are effective at introducing reactive oxygen species into the damaged polymer chain. These methods yield the lowest values of the water contact angle and form polar functional groups on the surface. Consequently, the dispersion component decreases, but the polar and dispersion components' combined effect leads to the highest surface free energy values among all activation methods.



	LP	CD	F	Pm	Cr	H	PA	I	O	K	P
γ_s [mJ/m ²]	49.0	44.3	39.37	35.9	31.7	31.2	28.4	27.5	27.3	25.1	22.1
γ_s^p [mJ/m ²]	26.7	22.9	20.46	1.7	3.7	0.9	14.1	2.0	0.6	3.5	0.9
γ_s^d [mJ/m ²]	22.3	21.4	18.91	34.2	28.0	30.3	14.3	25.5	26.7	21.6	21.2

Figure 3. Variation of the contact angle of water and diiodomethane and the components of the free surface energy after applying different treatment methods.

Chromium mixture (Cr) contributes less to the formation of polar functional groups on the surface of the tested samples. The simultaneous decrease in the dispersive component prevents this method from achieving high surface-free energy values. 240

The methods of applying primers (Pm, P) deserve attention. Both primers consist of modified polypropylenes, with Pm, being polypropylene grafted maleic anhydride and P being chlorinated polypropylene. These primers exhibit low wettability, as evidenced by the low polar component values obtained in the study. In the case of the primer based on polypropylene-grafted maleic anhydride (Pm), the high surface free energy results from strong dispersion interactions, although they are not as pronounced as in the primer based on chlorinated polypropylene (P). On the other hand, the water contact angle is higher even than for the substrate, which may be due to the addition of talc. 245 250

Despite having low polar component values, mechanical abrasion and immersion in chromium mixture significantly increase surface free energy compared to the non-activated sample. However, relative to the not-treated sample, the dispersive component increases by 7.5, 3.6, and 1.3 mJ/m² for Pm, H, and Cr, respectively. 255

In conclusion, dispersive interactions play a significant role in addition to the creation of polar functional groups observed in techniques such as CD, PA, LP, F, and Cr. These dispersive interactions can contribute to even higher surface free energy values than surface oxidizing methods, particularly for low-wetting polymers. 260

3.2. EDX study and SEM imaging results

SEM imaging was performed simultaneously with EDX analysis. SEM images were compared at x1000 magnification (Figure 5). At this resolution, changes in morphology after activation of the samples were most clearly visible. 265

In the EDX studies, the oxygen content indicated the presence of oxygen groups, while the number of impurities reflected the effectiveness of the washing process. The detected contaminants include all elements identified in the EDX analysis except oxygen and carbon. Figure 4 displays the X-ray spectrum obtained from the not-treated sample, which exhibited high impurity levels and low oxygen content. SEM images can confirm the presence of impurities (Figure 5). They are visible as irregular shapes in the SEM images, especially on the surface of the not-treated sample. Among the identified elements, apart from oxygen and carbon, the majority originated from silicon compounds ($8.0 \pm 0.9\%$), magnesium ($5.6 \pm 0.6\%$), and nitrogen ($5.0 \pm 1.2\%$). The concentration of the last element on the surface of the analyzed samples exhibited significant variability. The silicon mentioned above and magnesium elements come from the talc with which the polypropylene is filled. Talc 270 275 280

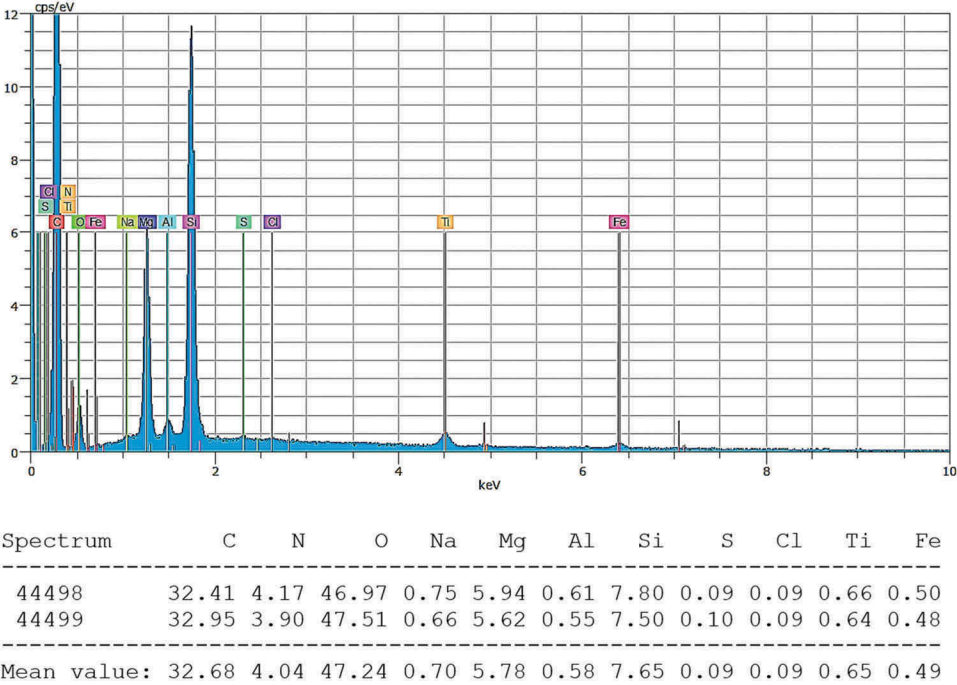


Figure 4. EDX spectra – sample 0.

particles can be shown in SEM images as small, white, oblong needles (Figure 5). Silicon and magnesium elements include inorganic impurities like salts, anti-adhesive coatings, and silica. However, due to the varying concentrations of these elements on the surface, they were considered as part of the surface contaminations.

The results of the EDX study are presented in Figure 6. Most of the mentioned pre-treatment techniques effectively removed surface contaminations, although several distinctions can be made. The surface treated with PA and F methods appeared to be the cleanest. The literature supports the high purification efficiency achieved through atmospheric plasma treatment, [45,46] whereas high temperatures can lead to surface degradation and oxidation. [47] This is evident from the increased oxygen content on the surface of samples treated with these methods. Additionally, samples treated with F, Cr, and LP activation showed elevated oxygen levels.

Among the physical surface treatment techniques, solvent cleaning and mechanical abrasion were the least effective in cleaning. It exhibited significant contamination, containing a substantial amount of nitrogen and other elements such as sodium, silicon, and magnesium. These impurities were also present in the not-treated sample (Figure 6) after the injection process. On the other hand, surfaces treated with mechanical abrasion showed contamination from particles in the emery paper, resulting in increased levels of silicon and

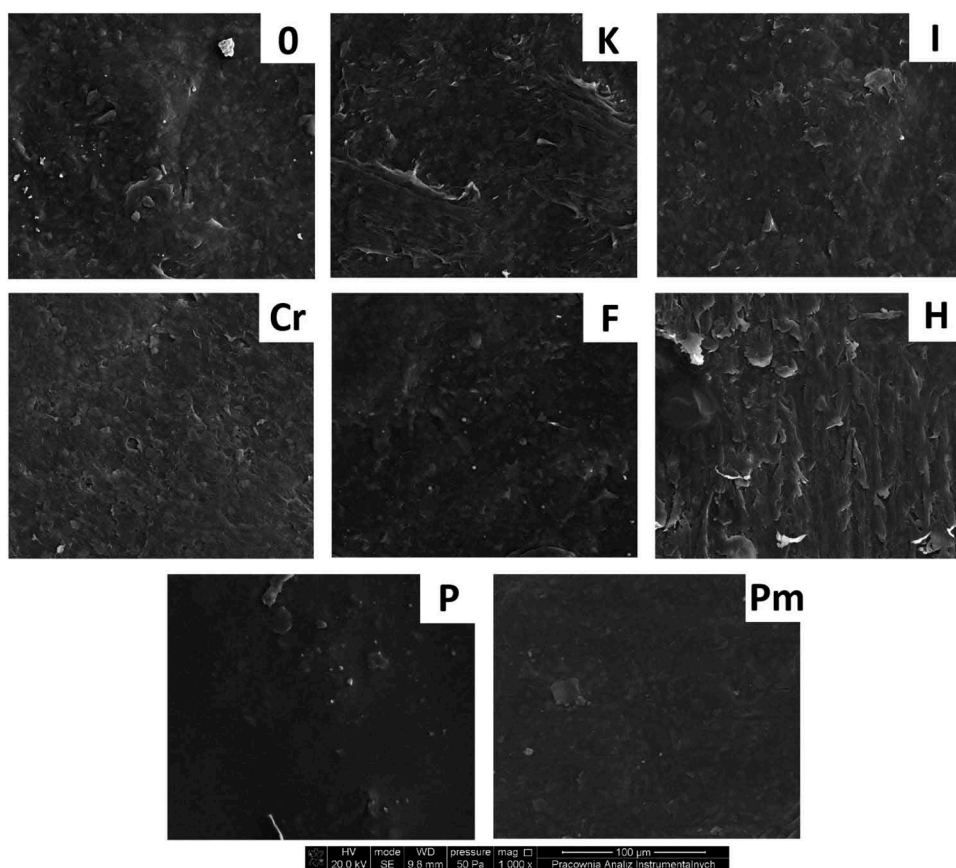


Figure 5. Scanning Electron Microscopy (SEM) results.

magnesium. At the same time, samples after washing with xylene and isopropanol have a lower silicon and magnesium content than the not-treated sample and the same as samples activated using other methods (F, PA, LP). This means that inorganic impurities that were not part of the talc have been removed, which is confirmed by SEM images (Figure 5).

The automatic CD and LP methods demonstrated relatively low cleaning efficiency in this study. However, the primary contamination on the surfaces treated with these techniques was nitrogen (3.5% and 4.1%, respectively). Apart from oxygen groups, nitrogen compounds can form during strong electrical discharges. After excluding the nitrogen content, CD and LP-treated surfaces contained low impurities, measuring 16.8% and 14.8%, respectively.

Among the methods that had minimal impact on the surface's oxygen content were solvents, mechanical abrasion, and corona discharges. It is known that immediately after corona treatment, hydrophobic recovery occurs for polyolefins.^[48,49]

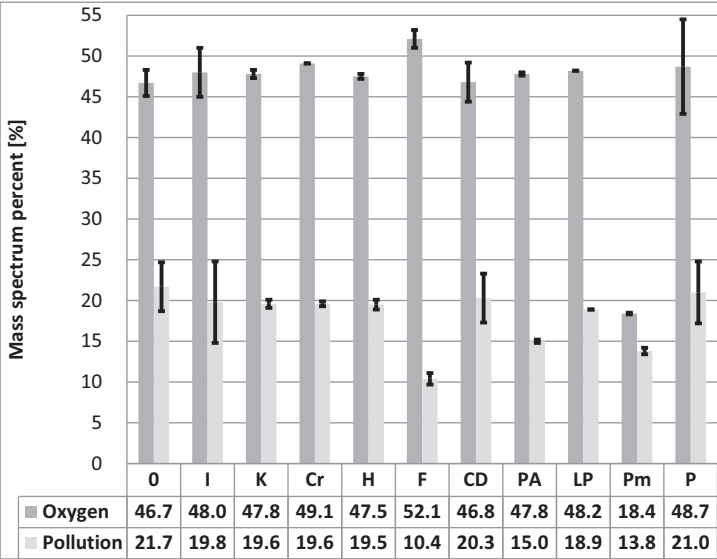


Figure 6. Contaminations (light gray) and amount of oxygen (dark gray) present on activated surfaces.

3.3. Atomic force (AFM) results

The same samples used for the EDX analysis were also subjected to the AFM and electron microscopy studies. The AFM results are presented as a graph (Figure 7).

Due to their specific car texture, the not-treated samples exhibited significant roughness. With polar functional groups and an oxidized surface (PA, LP, Cr, F, CD), increased roughness can enhance adhesion by increasing the contact area. However, adhesion is lower for non-polar surfaces (I, K, H, P, Pm) as the adhesive cannot penetrate the pores and cavities. In this test,

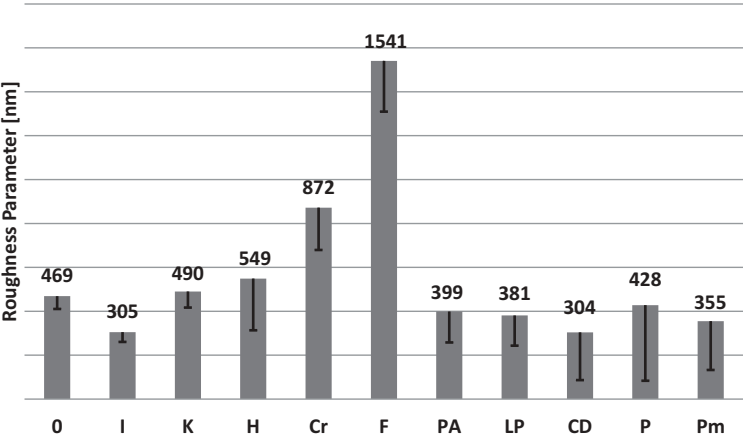


Figure 7. Surface roughness of polypropylene samples treated with various way.

adhesive tape is used, which reduces the mobility of the adhesive layer, resulting in reduced penetration of pores and cavities. For this reason, the surface of the sample should be smooth for maximum adhesion effect.

Roughness reduction was observed in almost all samples, with the most pronounced effect observed in the I and CD methods. This indicates the removal of surface contaminations present on the non-cleaned surface. Additionally, surface smoothing can occur due to the degradation of the roughest parts of the samples. The low roughness of the CD technique supports achieving high adhesion.

In this study, the specimens had automotive graining, resulting in significant roughness. In this case, mechanical abrasion (H samples) only slightly increased the roughness by unifying the height of the entire substrate. Samples abraded with sandpaper exhibit lamellar topography (Figure 5) with high numbers of ridges and valleys.

Compared to the non-treated sample, a significant increase in roughness was observed for the F and Cr methods, which is beneficial for highly polar surfaces after chromium etching. SEM images (Figure 5) show additional pores not visible in the not-treated sample. This proves that samples treated with flame and chromium mixture undergo strong oxidative etching. The porous structure can increase the adhesive properties in the case of similar polarity of the activated substrate and the adhesive. Hydrogen bonds attract the adhesive into the porous surface and becomes firmly anchored in the recesses.

The primers (P and Pm) create a layer that spreads to fill gaps and imitate the shape of the substrate. The roughness of both primer layers is slightly lower than that of the not-treated sample, confirming findings from other studies.^[50]

In the case of cleaning with isopropanol, the roughness decreased by 35%, whereas for xylene, the surface roughness remained almost the same as that of the not-treated sample. This suggests that after xylene washing, the polypropylene surface undergoes slight swelling. This hypothesis is supported by SEM images that show a partially dissolved and rubbed surface after xylene cleaning. Conversely, isopropanol, a commonly used solvent in industrial cleaning, effectively reduces impurities and smoothes the polypropylene surface.

3.4. Mechanical properties

The adhesive tape exhibited cohesive failures for all samples with a peel strength above 55 N/mm. Cohesive failures resulted in island break-offs or total failures, as depicted in Figure 8. The collected results were presented in a graph (Figure 9).

Peel strength testing is crucial in the automotive industry as it reflects the performance of adhesive bonding, where the adhesive joint is subjected to various load forces in different directions, including tension, shear, peel, and



Figure 8. Cohesive failure of the samples after the peel 180 strength test. On the left: failure observed in the form of islands. On the right: total cohesive failure.

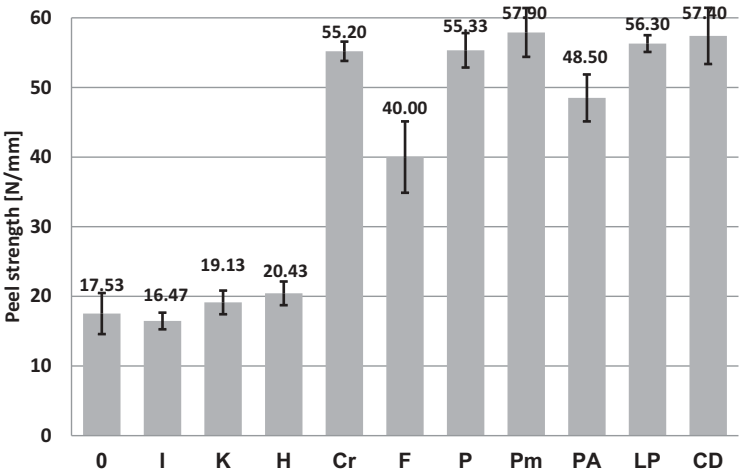


Figure 9. The results of peel strength tests (acc. ASTM D903–98). Samples after activation were joined with adhesive tape DuploCOLL 56,066.

torsion. Key adhesive tests used in the automotive industry are 180-degree peel and 90-degree tests. These stress scenarios can be observed, for instance, when peeling off a glass from a window frame or when a detachment of plastic covers and handles.

Significant improvements in the strength of bonded joints were observed with the Cr, LP, PA, CD, P, and Pm methods. A slight increase in peel strength was also noticed for the H and F methods. Other pre-treatment techniques did

not exhibit significant changes in peel strength compared to the not-treated sample. Beevers and Norris^[18] performed polypropylene surface activation before the bonding process using similar techniques as in this study. The activated samples were bonded to aluminium substrates using epoxy glue and tested using the shear strength of single-lap joints. In their tests, plasma activation (0.4 kN), flame treatment (0.4 kN), mechanical abrasion (0.6 kN), and wiping in solvent (0.1 kN) also did not achieve high results, while immersion in a chrome mixture (1.66 kN), primer application (1.48 kN) and corona discharge (1.27 kN) were among the best methods.

The automatic methods, particularly plasma activation,^[12–16,45–49,51] are widely recognized for enhancing peel adhesion strength, as confirmed by the conducted tests. The high peel strength of samples treated with LP, PA, and CD techniques and after immersion in a chromium mixture can be attributed to the heavily oxidized surface. Besides oxygen groups, nitrogen groups were also detected through EDX analysis. Introducing these groups requires cutting polymer chains and incorporating oxygen atoms into the damaged fragments. As a result of these processes, the surface of the polymer undergoes degradation, impacting the mechanical properties of the substrate and the subsequent environmental resistance of the adhesive joint.

The application of primers demonstrated exceptional efficiency in enhancing the adhesive bonding process. Ryntz^[52] showed that the solvent in the primer causes swelling of the top layer of polypropylene, resulting in a loosening of its polymeric structure. The functionalized polypropylene, which is the active substance of the primer, migrates in the loosened structure of the substrate. Due to the high similarity of the substrate – polypropylene to the primer active substance – chemically modified polypropylene, diffusion is promoted. Clemens^[24] reports that this substance diffuses up to 150 µm deep into polypropylene. Dispersion interactions via van der Waals interactions and mechanical interlocking lead to an interphase composed of primer and substrate material.^[53] After evaporation of the solvent, the intermolecular movements stop, and the interphase is stabilized, providing significant bond strength (Figure 9).

The limited increase in adhesion observed for other methods can be attributed to their surface-free energy, which did not reach a sufficient level. The energy generated by flame activation was insufficient to break the C-C bonds in polypropylene and introduce oxygen groups in this study. Conversely, mechanical abrasion significantly increased surface free energy due to dispersion interactions, resulting in higher strength test results.

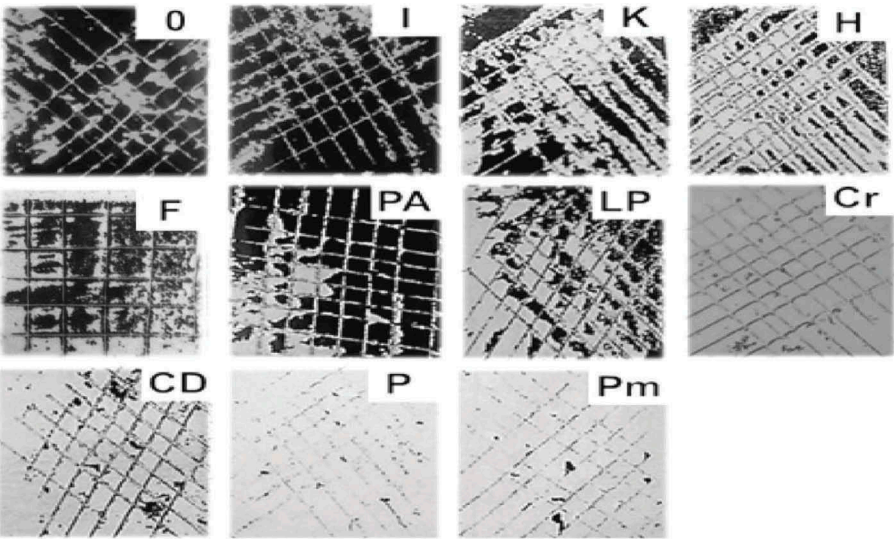
3.5. Paint coating adhesion test – cross-cut test

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The cross-cut test is a commonly used method in the automotive industry to quickly and intuitively assess the adhesion strength between the substrate and the applied paint layer, applicable to both polymer and metal substrates. It helps evaluate the effectiveness of the painting process.

This study performed the test twice, with two samples taken from each replicate. The sample with a larger amount of detached paint was selected for testing. The results are presented as photos of an adhesive tape with the paint detached from the substrate (Figure 10). The extent of delaminated paint indicates the level of adhesion to the substrate. Image Color Summarizer was used to calculate the amount of paint remaining on the adhesive tape.

These results were then classified according to ISO 2409^[41] standards. The results were color-coded to facilitate identification: green for good adhesion, orange for sufficient adhesion, and red for poor adhesion. Similar color codes are commonly used in the automotive industry to quickly verify results.



Sample	0	I	K	H	Cr	F	PA	LP	CD	P	Pm
Amount of detached paint [%]	66	54	38	26	10	57	56	43	16	2	3
ISO 2409 Classification	4	4	4	3	2	4	4	4	3	1	1

Figure 10. Photos of detached paint onto adhesive tape, amount of detached paint (%) and suitable classification according to ISO 2409.^[41]

In this study, the not-treated samples exhibited similar adhesive properties as those cleaned with isopropanol (I). This suggests that the polypropylene surface is clean after the injection process, which confirms the findings of the EDX analysis. 435

Some pre-treatment techniques, such as flame activation (F), atmospheric (PA), and low-pressure (LP) plasma, proved to be ineffective in improving paint adhesion. These techniques are generally known for their high efficiency in enhancing adhesion properties in the painting process.^[54–56] 440
However, various parameters, such as fuel feed amount, conveyor speed, process duration, and generator power, can influence the quality of the bond. Despite attempts to optimize these parameters, satisfactory results were not achieved.

Surprisingly, both xylene cleaning and mechanical abrasion did not enhance adhesion in the adhesive bonding process but showed some efficiency 445
in painting. Xylene, being an organic non-polar solvent, shares chemical similarities with polypropylene and paint compositions, where it is commonly used as a primary solvent. It functions through wetting and slight substrate dissolution, as observed in SEM images (Figure 5). Conversely, mechanical abrasion results in a smoother surface, providing a larger contact area between 450
the substrate and the applied paint layer in case of ineffective wetting.

Chemical methods, including immersion in a chromic acid mixture and application of primers, demonstrated the highest efficiency in painting. The immersion process contributes to surface damage and introduces a significant number of oxygen groups on the polypropylene surface. Primers, on the other 455
hand, do not degrade the substrate; instead, it strengthens the adhesive joint by providing an additional protective layer on the polypropylene. The efficiency of painting significantly improves regardless of the type of primer used.

4. Summary and conclusions

The polypropylene substrates, which acquired a characteristic car texture, 460
underwent activation using ten pre-treatment methods. Each method had a distinct impact on the polypropylene surface, such as reducing or increasing roughness, introducing polar functional groups, removing a weak boundary layer, or promoting polymer chains diffusion to create a stable interphase.

The effectiveness of these techniques was evaluated using six research 465
methods, two of which were practical methods commonly used to assess the strength of adhesive joints in the automotive industry. A summary of all the studies is presented in Table 3. Positive results are indicated by a green (+), sufficient results by an orange ($\pm$), and insufficient results by a red (–). These indications refer to the outcomes obtained for each specific surface treatment 470
method compared to the not-treated sample.

Table 3. The obtained results summary. Green (+) indicates an increase in the current feature compared to the not-treated sample (or a decrease in the case of contamination and roughness). Orange (±) represents a slight increase (or slight reduction in the case of roughness). Red (–) indicates a decrease in the current feature (or an increase in the case of contamination and roughness).

Parameter	0	I	K	Cr	H	F	CD	PA	LP	Pm	P
Surface Free Energy	–	–	–	+	+	+	+	±	+	+	–
Surface oxygen	–	±	±	+	±	+	–	±	+	–	+
Contamination removing	–	+	+	+	+	+	±	+	+	+	±
Surface roughness	–	+	±	–	–	–	+	+	+	+	+
Adhesive bonding	–	–	–	+	±	±	+	+	+	+	+
Painting	–	–	–	+	±	–	±	–	–	+	+

Several studied techniques have proven effective in increasing bondability. However, exposition of these joints for climatic conditions in aging tests could provide further information on which of these methods show resistance to complex environmental conditions.

The study revealed that samples activated using automatic methods (F, PA, LP, DC) exhibited significant surface oxidation, increasing the polar component of surface free energy and the strength of adhesive joints. However, this activation method also caused damage to the material structure, which could be of significant importance, especially for thinner fragments of automotive parts. In the presence of strong electrical discharges, not only oxygen-functional groups but also nitrogen-containing groups were formed. Flame activation is particularly susceptible to changes in process parameters. The selection of appropriate parameters can drastically alter the adhesion of the applied coating to the substrate.

Furthermore, we demonstrated that adhesive coatings, known as primers, can effectively enhance adhesion properties and compete with well-established surface activation methods. Primers showed efficiency in both adhesive bonding and painting processes. The adhesion-increasing effect is attributed to the dispersion interactions dominating during the intertwining of the primer and substrate polymer chains.^[26,27] SEM photos and EDX results also show that the primers can cover surface irregularities and unremoved contaminants. This effectiveness was confirmed using a commercial primer based on chlorinated polypropylene (P) and a self-made primer containing polypropylene grafted with maleic anhydride (Pm). The primer method in the manual application does not require expensive machinery or extensive parameter settings and does not damage the activated surface. They are used in very

diminutive volume, on a small surface subjected to adhesive bonding or painting, and therefore do not often require an extractor. They can be successfully applied on any production scale, and after optimization of parameters also in automatic lines. 500

Primers based on chlorinated ingredients are very common in the industry. In our research, we have shown that the use of a non-chlorinated primer can produce similar adhesion results while reducing the amount of chlorinated substances released into the environment. 505

Although immersion in a chromium mixture yielded promising results in adhesion tests, it was observed that this activation method damages the surface and alters the original appearance and geometry of the activated samples. Moreover, substances used in the chromium mixture have carcinogenic effects, leading to its withdrawal from industrial practice. 510

Several methods that reduced substrate roughness also demonstrated improved adhesion properties. The techniques that exhibited this effect were corona discharges (CD), atmospheric (PA), and low-pressure (LP) plasma, as well as the use of primers (P, Pm). In the case of a rigid adhesive bond such as adhesive tape, the smooth surface promotes adhesion properties. 515

Two solvents, isopropanol (I) and xylene, were employed for washing the polypropylene surface. The EDX study indicated that both solvents were equally effective in cleaning. However, xylene resulted in a rougher surface, whereas isopropanol had the opposite effect. The increased surface roughness and substrate swelling after cleaning with xylene led to enhanced adhesion properties in the cut and peel tests compared to the samples cleaned with isopropanol. 520

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